SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2- INDUSTRY PROBLEM SOLVING

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4 - Industry Problem Solving	Assessment:	ASSESSMENT TOOL 2- INDUSTRY PROBLEM SOLVING
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:	Jeya Harshini R	Reg. No.:	192424051
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Code of Conduct

I Jeya Harshini R (192424051) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Jeya Harshini R

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

		reasoning and insights				
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Question 1 — E-Commerce

Problem Statement

An online retailer has **50,000 customers**. A random sample of **500 customers** shows that **68% made a purchase** during a promotional campaign. Estimate the proportion of all customers who made a purchase.

---

1. Identify the Population and Sample

Population:

The entire group of **50,000 customers** of the online retailer.

Sample:

A randomly selected group of **500 customers**.

Given:

$$N = 50,000$$

$$n = 500$$

Sample proportion:

$$\hat{p} = 0.68$$

Therefore, the sample contains:

$$500(0.68) = 340$$

customers who made a purchase.

---

2. Identify the Parameter of Interest

The parameter of interest is the **population proportion** of all customers who made a purchase during the promotional campaign.

Let the population proportion be represented by:

$$p$$

The objective is to estimate (p).

---

3. Calculate the Point Estimate

The point estimate of a population proportion is the sample proportion.

Formula:

$$\hat{p} = \frac{x}{n}$$

where:

- (x) = number of customers who made a purchase
- (n) = sample size

Here,

$$x = 340$$

and

$$n = 500$$

Therefore,

$$\hat{p} = \frac{340}{500}$$
$$\hat{p} = 0.68$$

In percentage form:

$$0.68 \times 100 = 68\%$$

Point Estimate = 0.68 or 68%

Thus, the estimated proportion of all customers who made a purchase is **68%**.

---

4. State the Null Hypothesis (H_0)

Since the problem does not provide a previously claimed population proportion for comparison, we can use the estimated value as the reference value for demonstrating the hypothesis-testing procedure.

$$H_0 : p = 0.68$$

The null hypothesis states that the population proportion of customers who made a purchase is **68%**.

---

5. State the Alternative Hypothesis (H_1)

Since we want to determine whether the actual population proportion differs from 68%, a two-tailed alternative hypothesis is appropriate.

$$H_1 : p \neq 0.68$$

The alternative hypothesis states that the actual population proportion is **different from 68%**.

---

6. Select the Significance Level (\alpha)

A standard significance level of:

$$\alpha = 0.05$$

is selected.

Therefore, the test is conducted at the **5% significance level**.

---

7. Calculate the Confidence Interval and Test Statistic

95% Confidence Interval

For a population proportion, the confidence interval is:

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

For a 95% confidence level:

$$z_{\alpha/2} = 1.96$$

Substituting the values:

$$0.68 \pm 1.96\sqrt{\frac{0.68(1-0.68)}{500}}$$

$$= 0.68 \pm 1.96\sqrt{\frac{0.68(0.32)}{500}}$$

$$= 0.68 \pm 1.96\sqrt{\frac{0.2176}{500}}$$

$$= 0.68 \pm 1.96\sqrt{0.0004352}$$

$$= 0.68 \pm 1.96(0.02086)$$

$$= 0.68 \pm 0.04089$$

Therefore,

$$Lower\ Limit = 0.68 - 0.04089$$

$$Lower\ Limit = 0.63911$$

and

$$Upper\ Limit = 0.68 + 0.04089$$

$$Upper\ Limit = 0.72089$$

95% Confidence Interval = (0.6391, 0.7209)

In percentage form:

95% CI = 63.91% to 72.09%

Test Statistic

For the hypothesis (H_0:p=0.68):

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

Here,

$$\hat{p} = 0.68$$

and

$$p_0 = 0.68$$

Therefore,

$$z = \frac{0.68 - 0.68}{\sqrt{\frac{0.68(0.32)}{500}}}$$

$$z = 0$$

Test Statistic = 0

---

8. Calculate the p-value

Since the test statistic is:

$$z = 0$$

the two-tailed p-value is:

$$p\text{-value} = 0.50$$

p-value = 0.50

---

9. Compare p-value with (\alpha)

Given:

$$p\text{-value} = 0.50$$

and

$$\alpha = 0.05$$

Since:

$$0.50 > 0.05$$

we **fail to reject the null hypothesis**.

---

10. Industry-Oriented Conclusion

The sample indicates that **68% of customers made a purchase** during the promotional campaign. Therefore, the point estimate of the population purchase proportion is **68%**.

The calculated 95% confidence interval ranges from approximately **63.91% to 72.09%**. This means that, based on the sample, the retailer can reasonably estimate that the true proportion of customers who made a purchase lies within this range.

The hypothesis test gives a p-value of **0.50**, which is greater than the significance level of **0.05**. Therefore, there is insufficient evidence to conclude that the population purchase proportion differs from 68%.

Final Result

Measure	**Result**
Population	50,000 customers
Sample	500 customers
Customers who purchased	340
Point Estimate	0.68
Point Estimate (%)	68%

Measure	Result
95% Confidence Interval	0.6391 – 0.7209
Confidence Interval (%)	63.91% – 72.09%
Significance Level	0.05
Test Statistic	0
p-value	0.50
Decision	Fail to reject (H_0)

Therefore, the estimated proportion of customers who made a purchase is 68%, with a 95% confidence interval of approximately 63.91%–72.09%.

Question 2 — Manufacturing

Problem Statement

A factory produces electronic components. From a random sample of **100 components**, the average lifetime is **1,250 hours**. Determine the point estimate of the population mean lifetime.

---

1. Identify the Population and Sample

Population:

The entire population of electronic components produced by the factory.

Sample:

A randomly selected sample of **100 electronic components**.

Given:

$$n = 100$$

$$\bar{x} = 1250 \text{ hours}$$

---

2. Identify the Parameter of Interest

The parameter of interest is the **population mean lifetime** of all electronic components produced by the factory.

Let the population mean be represented by:

$$\mu$$

The objective is to estimate the true value of (\mu).

---

3. Calculate the Point Estimate

The point estimate of the population mean is the **sample mean**.

Formula:

$$\hat{\mu} = \bar{x}$$

Given:

$$\bar{x} = 1250$$

Therefore:

$$\boxed{\hat{\mu} = 1250 \text{ hours}}$$

Point Estimate = 1,250 hours

This means that the estimated average lifetime of all electronic components produced by the factory is **1,250 hours**.

---

4. State the Null Hypothesis (H_0)

Since the problem does not provide a previously established or claimed population mean, we use the sample estimate as the reference value for demonstrating the hypothesis-testing procedure.

$$H_0 : \mu = 1250$$

The null hypothesis states that the true average lifetime of the electronic components is **1,250 hours**.

---

5. State the Alternative Hypothesis (H_1)

To determine whether the true population mean differs from 1,250 hours:

$$H_1 : \mu \neq 1250$$

The alternative hypothesis states that the true average lifetime of the components is **different from 1,250 hours**.

---

6. Select the Significance Level (\alpha)

A standard significance level of:

$$\alpha = 0.05$$

is selected.

Therefore, the hypothesis test is conducted at the **5% significance level**.

---

7. Calculate the Confidence Interval and/or Test Statistic

The question provides the **sample mean and sample size**, but it does **not provide the sample standard deviation or population standard deviation**.

Therefore, an exact numerical confidence interval or test statistic **cannot be calculated from the given information**.

The general 95% confidence interval for the population mean, when the population standard deviation is unknown, is:

$$\bar{x} \pm t_{\alpha/2,n-1}\frac{s}{\sqrt{n}}$$

where:

- $\bar{x}$ = sample mean
- s = sample standard deviation
- n = sample size
- $t_{\alpha/2,n-1}$ = t-critical value

For this problem:

$$\bar{x} = 1250$$

$$n = 100$$

Therefore:

$$CI = 1250 \pm t_{0.025,99}\frac{s}{\sqrt{100}}$$

$$CI = 1250 \pm t_{0.025,99}\frac{s}{10}$$

A numerical confidence interval requires the value of (s).

Test Statistic

The one-sample t-test statistic is:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Using:

$$\bar{x} = 1250$$

and

$$\mu_0 = 1250$$

we get:

$$t = \frac{1250 - 1250}{s/\sqrt{100}}$$

$$t = 0$$

Test Statistic = 0

However, calculation of the exact p-value requires the sample standard deviation.

---

8. Calculate the p-value

For the demonstration hypothesis:

$$H_0 : \mu = 1250$$

the calculated test statistic is:

$$t = 0$$

For a two-tailed test, a test statistic of zero corresponds to:

$$p\text{-value} = 1.00$$

p-value = 1.00

---

9. Compare p-value with (\alpha)

Given:

$$p\text{-value} = 1.00$$

and:

$$\alpha = 0.05$$

Since:

$$1.00 > 0.05$$

we **fail to reject** H_0.

---

10. Industry-Oriented Conclusion

The sample of 100 electronic components has an average lifetime of **1,250 hours**. Therefore, the **point estimate of the population mean lifetime is 1,250 hours**.

This estimate indicates that the factory's electronic components are expected to have an average lifetime of approximately **1,250 hours**.

For the illustrative hypothesis test, the test statistic is 0 and the p-value is 1.00, which is greater than the significance level of 0.05. Hence, there is insufficient evidence to reject the null hypothesis that the population mean is 1,250 hours.

However, the problem does not provide the sample standard deviation. Therefore, a numerical confidence interval cannot be completely calculated from the given data.

Final Result

Measure	Result
Population	All electronic components produced by the factory
Sample	100 components
Sample Mean	1,250 hours

Measure	Result
Point Estimate of Population Mean	**1,250 hours**
Significance Level	0.05
Test Statistic	0
p-value	1.00
Decision	Fail to reject (H_0)
95% Confidence Interval	Cannot be numerically determined without sample SD

Final Answer

$$\boxed{\hat{\mu} = \bar{x} = 1250 \text{ hours}}$$

Therefore, the point estimate of the population mean lifetime of the electronic components is 1,250 hours.

Question 3 — Banking

Problem Statement

A bank wants to estimate the average monthly transaction value of its customers. A sample of **200 customers** has a mean transaction value of **₹18,500** and standard deviation of **₹4,200**. Calculate the point estimates of the population mean and variance.

---

1. Identify the Population and Sample

Population:

All customers of the bank whose monthly transaction values are being studied.

Sample:

A randomly selected sample of **200 customers**.

Given:

$$n = 200$$

$$\bar{x} = ₹18,500$$

$$s = ₹4,200$$

---

2. Identify the Parameters of Interest

There are two parameters of interest:

1. **Population mean transaction value**, represented by (\mu).
2. **Population variance of transaction values**, represented by (\sigma^2).

The objective is to estimate both parameters using the sample data.

---

3. Calculate the Point Estimates

(a) Point Estimate of Population Mean

The point estimate of the population mean is the **sample mean**.

Formula:

$$\hat{\mu} = \bar{x}$$

Given:

$$\bar{x} = ₹18,500$$

Therefore:

$$\boxed{\hat{\mu} = ₹18,500}$$

Estimated Population Mean = ₹18,500

This means that the estimated average monthly transaction value of all bank customers is **₹18,500**.

(b) Point Estimate of Population Variance

The sample variance is used as the point estimate of the population variance.

Formula:

$$\hat{\sigma}^2 = s^2$$

Given:

$$s = ₹4,200$$

Therefore:

$$s^2 = (4200)^2$$

$$s^2 = 17,640,000$$

Hence:

$$\boxed{\hat{\sigma}^2 = ₹17,640,000}$$

The units of variance are **rupees squared ($₹^2$)**.

Estimated Population Variance = 17,640,000 $₹^2$

---

4. State the Null Hypothesis (H_0)

Since the problem does not provide an existing population value for comparison, we use the calculated point estimate as the reference value for demonstrating the hypothesis-testing procedure.

For the population mean:

$$H_0 : \mu = ₹18,500$$

The null hypothesis states that the true average monthly transaction value is **₹18,500**.

For the population variance:

$$H_0 : \sigma^2 = 17,640,000$$

The null hypothesis states that the population variance is **17,640,000 ₹²**.

---

5. State the Alternative Hypothesis (H_1)

For the population mean:

$$H_1 : \mu \neq ₹18,500$$

This means that the actual population mean transaction value is different from ₹18,500.

For the population variance:

$$H_0 : \sigma^2 = 17,640,000$$

This means that the actual population variance differs from 17,640,000 ₹².

---

6. Select the Significance Level (\alpha)

A standard significance level of:

$$\alpha = 0.05$$

is selected.

Therefore, the hypothesis tests are conducted at the **5% significance level**.

---

7. Calculate the Confidence Interval and/or Test Statistic

Confidence Interval for Population Mean

Since the sample size is:

$$n = 200$$

which is greater than 30, a large-sample confidence interval can be calculated using the normal approximation.

Formula:

$$CI = \bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}$$

For a 95% confidence level:

$$z_{\alpha/2} = 1.96$$

Substituting:

$$CI = 18500 \pm 1.96 \frac{4200}{\sqrt{200}}$$

$$\sqrt{200} = 14.142$$

Therefore:

$$CI = 18500 \pm 1.96 \left(\frac{4200}{14.142} \right)$$

$$CI = 18500 \pm 1.96(296.98)$$

$$CI = 18500 \pm 582.08$$

Therefore:

$$Lower\ Limit = 18500 - 582.08$$

$$= 17917.92$$

and:

$$Upper\ Limit = 18500 + 582.08$$

$$= 19082.08$$

95% Confidence Interval = ₹17,917.92 to ₹19,082.08

Test Statistic for Population Mean

Using:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Here:

$$\bar{x} = 18500$$

and:

$$\mu_0 = 18500$$

Therefore:

$$t = \frac{18500 - 18500}{4200/\sqrt{200}}$$

$$t = 0$$

Test Statistic = 0

---

8. Calculate the p-value

For a two-tailed test with:

$$t = 0$$

the p-value is:

$$\boxed{p\text{-value} = 1.00}$$

---

9. Compare p-value with (\alpha)

Given:

$$p\text{-value} = 1.00$$

$$\alpha = 0.05$$

Since:

$$1.00 > 0.05$$

we **fail to reject the null hypothesis**.

---

10. Industry-Oriented Conclusion

The sample of 200 customers has an average monthly transaction value of **₹18,500** and a standard deviation of **₹4,200**.

The point estimate of the population mean is therefore:

$$\boxed{₹18,500}$$

The point estimate of the population variance is:

$$\boxed{17,640,000\ ₹^2}$$

The 95% confidence interval for the population mean is approximately:

$$\boxed{₹17,917.92 \text{ to } ₹19,082.08}$$

This indicates that the bank can reasonably estimate the true average monthly transaction value of its customer population to lie within this range.

The illustrative hypothesis test produces a p-value of 1.00, which is greater than the significance level of 0.05. Therefore, there is insufficient evidence to reject the null hypothesis that the population mean is ₹18,500.

From a banking perspective, this information can help the bank understand typical customer transaction behavior, design suitable banking products, identify customer segments, and plan transaction-related services.

Final Result

Measure	Result
Population	All bank customers
Sample	200 customers
Sample Mean	₹18,500
Sample Standard Deviation	₹4,200
Point Estimate of Population Mean	**₹18,500**
Point Estimate of Population Variance	**17,640,000 ₹^{2}**
95% Confidence Interval	**₹17,917.92 – ₹19,082.08**
Significance Level	0.05
Test Statistic	0
p-value	1.00
Decision	Fail to reject (H_0)

Final Answer

$$\boxed{\hat{\mu} = ₹18,500}$$

$$\boxed{\hat{\sigma}^2 = 17,640,000\ ₹^2}$$

Therefore, the point estimate of the population mean monthly transaction value is ₹18,500, and the point estimate of the population variance is 17,640,000 ₹^{2}.

Question 4 — Healthcare Analytics

Problem Statement

A hospital randomly selects **400 patients**, of whom **52 experienced a particular side effect**. Find the point estimate of the proportion of patients experiencing the side effect. Explain what the estimate means.

---

1. Identify the Population and Sample

Population

The population consists of **all patients who receive the treatment or medication being studied by the hospital**.

Sample

The hospital randomly selects **400 patients** for the study.

Given:

$$n = 400$$

Number of patients who experienced the side effect:

$$x = 52$$

Number of patients who did not experience the side effect:

$$400 - 52 = 348$$

Therefore:

- **Sample size = 400 patients**

- **Patients with side effect = 52**
- **Patients without side effect = 348**

---

2. Identify the Parameter of Interest

The parameter of interest is the **population proportion of patients who experience the particular side effect**.

Let the population proportion be represented by:

$$p$$

The objective is to estimate the true value of (p) using the sample data.

---

3. Calculate the Point Estimate

The point estimate of a population proportion is the **sample proportion**.

Formula:

$$\hat{p} = \frac{x}{n}$$

Substituting the given values:

$$\hat{p} = \frac{52}{400}$$

$$\hat{p} = 0.13$$

Converting into percentage:

$$0.13 \times 100 = 13\%$$

Point Estimate = 0.13 or 13%

Therefore, the estimated proportion of patients experiencing the side effect is:

$$\boxed{13\%}$$

---

4. State the Null Hypothesis (H_0)

The question does not provide a previously established population proportion for comparison. Therefore, for demonstrating the hypothesis-testing procedure, the estimated proportion is used as the reference value.

$$H_0 : p = 0.13$$

The null hypothesis states that the true population proportion of patients experiencing the side effect is **13%**.

---

5. State the Alternative Hypothesis (H_1)

Since we are checking whether the actual population proportion differs from 13%, a two-tailed hypothesis is considered.

$$H_1 : p \neq 0.13$$

The alternative hypothesis states that the true population proportion of patients experiencing the side effect is **different from 13%**.

---

6. Select the Significance Level (\alpha)

A standard significance level is selected:

$$\alpha = 0.05$$

Therefore, the hypothesis test is conducted at the **5% significance level**.

---

7. Calculate the Confidence Interval and Test Statistic

95% Confidence Interval

For a population proportion, the confidence interval is calculated using:

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

For a 95% confidence level:

$$z_{\alpha/2} = 1.96$$

Given:

$$\hat{p} = 0.13$$

$$n = 400$$

Substituting:

$$\begin{aligned} CI &= 0.13 \pm 1.96\sqrt{\frac{0.13(1-0.13)}{400}} \\ &= 0.13 \pm 1.96\sqrt{\frac{0.13(0.87)}{400}} \\ &= 0.13 \pm 1.96\sqrt{\frac{0.1131}{400}} \\ &= 0.13 \pm 1.96\sqrt{0.00028275} \\ &= 0.13 \pm 1.96(0.01682) \\ &= 0.13 \pm 0.03297 \end{aligned}$$

Therefore, the lower limit is:

$$0.13 - 0.03297 = 0.09703$$

The upper limit is:

$$0.13 + 0.03297 = 0.16297$$

95% Confidence Interval

$$\boxed{(0.0970,\ 0.1630)}$$

In percentage form:

$$\boxed{9.70\% \text{ to } 16.30\%}$$

Test Statistic

For the hypothesis:

$$H_0 : p = 0.13$$

the one-proportion z-test statistic is:

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

Here:

$$\hat{p} = 0.13$$

$$p_0 = 0.13$$

Therefore:

$$z = \frac{0.13 - 0.13}{\sqrt{\frac{0.13(0.87)}{400}}}$$

$$z = 0$$

Test Statistic = 0

---

8. Calculate the p-value

For a two-tailed test with:

$$z = 0$$

the p-value is:

$$\boxed{p\text{-value} = 1.00}$$

---

9. Compare p-value with (\alpha)

Given:

$$p\text{-value} = 1.00$$

and:

$$\alpha = 0.05$$

Comparison:

$$1.00 > 0.05$$

Therefore, we **fail to reject the null hypothesis**.

---

10. Industry-Oriented Conclusion

The hospital studied a random sample of **400 patients**, out of which **52 patients experienced the particular side effect**.

The estimated proportion is:

$$\hat{p} = 0.13$$

Therefore, approximately **13% of patients** in the population are estimated to experience the side effect.

The 95% confidence interval ranges from approximately:

$$9.70\% \text{ to } 16.30\%$$

This means that the hospital can reasonably estimate that the true proportion of patients experiencing the side effect lies between approximately **9.70% and 16.30%**.

From a healthcare analytics perspective, this estimate can help the hospital monitor treatment safety, identify potential adverse effects, inform clinical decisions, and support patient risk assessment.

The illustrative hypothesis test gives a p-value of **1.00**, which is greater than the significance level of **0.05**. Hence, there is insufficient evidence to reject the null hypothesis that the population proportion is 13%.

Final Result

Measure	**Result**
Population	All patients receiving the treatment
Sample	400 patients
Patients experiencing side effect	52
Patients without side effect	348
Point Estimate	**0.13**
Point Estimate (%)	**13%**
95% Confidence Interval	**0.0970 – 0.1630**
Confidence Interval (%)	**9.70% – 16.30%**
Significance Level	**0.05**
Test Statistic	**0**
p-value	**1.00**
Decision	**Fail to reject (H_0)**

Final Answer

$$\boxed{\hat{p} = \frac{52}{400} = 0.13}$$

$$\boxed{\hat{p} = 13\%}$$

Therefore, the point estimate of the proportion of patients experiencing the particular side effect is 13%. The 95% confidence interval is approximately 9.70% to 16.30%.

Question 5 — Food Industry

Problem Statement

A food-processing company measures the weight of **64 randomly selected packets**. The mean weight is **498 g** and the sample standard deviation is **12 g**. Construct a **95% confidence interval** for the true average packet weight.

---

1. Identify the Population and Sample

Population

The population consists of **all packets produced by the food-processing company**.

Sample

A random sample of **64 packets** is selected for quality inspection.

Given:

$$n = 64$$

$$\bar{x} = 498 \text{ g}$$

$$s = 12 \text{ g}$$

---

2. Identify the Parameter of Interest

The parameter of interest is the **population mean packet weight**.

Let:

$$\mu$$

represent the true average packet weight of all packets produced by the company.

---

3. Calculate the Point Estimate

The point estimate of the population mean is the sample mean.

Formula:

$$\hat{\mu} = \bar{x}$$

Substituting:

$$\hat{\mu} = 498$$

Point Estimate = 498 g

Therefore, the estimated average packet weight is:

$$\boxed{498 \text{ g}}$$

---

4. State the Null Hypothesis (H_0)

The standard packet weight claimed by the company is assumed to be **500 g**.

$$H_0 : \mu = 500$$

The null hypothesis states that the true average packet weight is **500 g**.

---

5. State the Alternative Hypothesis (H_1)

$$H_1 : \mu \neq 500$$

The alternative hypothesis states that the true average packet weight is **different from 500 g.**

---

6. Select the Significance Level (\alpha)

A 95% confidence level corresponds to:

$$\alpha = 0.05$$

Therefore:

$$\boxed{\alpha = 0.05}$$

---

7. Calculate the Confidence Interval and Test Statistic

Step 1: Calculate Standard Error

Formula:

$$SE = \frac{s}{\sqrt{n}}$$

Substituting:

$$SE = \frac{12}{\sqrt{64}}$$

$$SE = \frac{12}{8}$$

$$SE = 1.5$$

Step 2: Determine Critical Value

For a 95% confidence level:

$$z_{0.025} = 1.96$$

Step 3: Calculate Margin of Error

Formula:

$$ME = z\left(\frac{s}{\sqrt{n}}\right)$$

$$ME = 1.96(1.5)$$

$$ME = 2.94$$

Step 4: Calculate Confidence Interval

Formula:

$$CI = \bar{x} \pm ME$$

$$CI = 498 \pm 2.94$$

Lower Limit:

$$498 - 2.94 = 495.06$$

Upper Limit:

$$498 + 2.94 = 500.94$$

95% Confidence Interval

$$\boxed{(495.06 \text{ g},\ 500.94 \text{ g})}$$

Test Statistic

Formula:

$$z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Substituting:

$$z = \frac{498 - 500}{12/\sqrt{64}}$$

$$z = \frac{-2}{1.5}$$

$$z = -1.33$$

Test Statistic = -1.33

---

8. Calculate the p-value

For:

$$z = -1.33$$

Two-tailed p-value:

$$p\text{-value} = 2P(Z < -1.33)$$

From the standard normal table:

$$P(Z < -1.33) = 0.0918$$

Therefore:

$$p\text{-value} = 2(0.0918)$$

$$p\text{-value} = 0.1836$$

p-value = 0.1836

---

9. Compare p-value with (\alpha)

Given:

$$p\text{-value} = 0.1836$$

$$\alpha = 0.05$$

Since:

$$0.1836 > 0.05$$

we **fail to reject the null hypothesis**.

---

10. Industry-Oriented Conclusion

The food-processing company selected **64 packets** and found an average packet weight of **498 g** with a sample standard deviation of **12 g**.

The point estimate of the population mean packet weight is:

$$\boxed{498 \text{ g}}$$

The 95% confidence interval for the true average packet weight is:

$$\boxed{495.06 \text{ g to } 500.94 \text{ g}}$$

This means that the company can be 95% confident that the true average weight of all packets produced lies between **495.06 g and 500.94 g**.

The hypothesis test produced:

$$z = -1.33$$

and

$$p\text{-value} = 0.1836$$

Since the p-value is greater than 0.05, there is insufficient evidence to conclude that the average packet weight differs significantly from the claimed weight of 500 g.

From a food industry quality-control perspective, the production process appears to be operating within acceptable limits, and there is no statistical evidence of underfilling or overfilling.

Final Result

Measure	**Result**
Population	All packets produced by the company
Sample	64 packets
Sample Mean	498 g
Sample Standard Deviation	12 g
Point Estimate of Population Mean	**498 g**
Standard Error	**1.5 g**
Margin of Error	**2.94 g**
95% Confidence Interval	**495.06 g – 500.94 g**
Significance Level	**0.05**
Test Statistic	**-1.33**
p-value	**0.1836**
Decision	**Fail to Reject (H_0)**

Final Answer

$$\boxed{\hat{\mu} = 498 \text{ g}}$$

$$\boxed{95\% \, CI = (495.06 \text{ g}, \ 500.94 \text{ g})}$$

$$\boxed{p\text{-value} = 0.1836}$$

Therefore, the estimated average packet weight is 498 g, and the 95% confidence interval for the true average packet weight is 495.06 g to 500.94 g. Since the p-value is greater than 0.05, there is no significant evidence that the true mean weight differs from 500 g.